MANIKANDAN SOMU

Geodesy & Geospatial Data Specialist
Remote Sensing — Climate Services — Earth Observation Engineer

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SUMMARY

Geospatial Engineer and Earth Observation specialist with professional experience in geospatial analysis, remote sensing, GIS, and environmental data processing. Recently completed an M.Sc. in Geodetic Engineering at the University of Bonn, with research focused on satellite-based environmental monitoring and water resource analysis. Experienced in working with Earth observation, climate, and geospatial datasets including GRACE/GRACE-FO, ERA5, GLDAS, satellite altimetry, and spatial databases. Skilled in Python, MATLAB, QGIS, geospatial workflow development, data visualization, and scientific analysis. Interested in Earth observation services, climate resilience, geospatial data architecture, environmental monitoring, and translating complex geospatial information into practical solutions for decision-making and climate action.

SKILLS

Programming & Development: Python, MATLAB, SQL, HTML, JavaScript

GIS & spatial Technologies: ESA Snap,QGIS, ArcGIS, PyQGIS, GDAL,PostgreSQl/PostGIS, OpenStreetMap

Earth Observation & Remote Sensing: GRACE/GRACE-FO, Landsat, ERA5, GLDAS, Satellite Altimetry,Copernicus Sentinel-1,2

Spatial Analysis & Processing: Raster/vector processing, Time-series analysis, Environmental Monitoring, geospatial workflows

Tools:REST APIs, Google Earth Engine

EDUCATION

University of Bonn — Bonn, Germany
M.Sc. in Geodetic Engineering — **Oct 2022 - Jun 2026**

College of Engineering Guindy, Anna University — Chennai, India
Bachelor's in Geoinformatics Engineering — **Aug 2015 - Jun 2019**

WORK EXPERIENCE

Working Student – Full Stack Geospatial Developer — **Jun 2024 - Jul 2025**
Navilan Solution GmbH — Bonn, Germany

- Developed full-stack geospatial applications using Python, web technologies, and spatial databases.
- Built and maintained backend services, APIs, and geospatial data processing workflows.
- Developed interactive WebGIS interfaces and geospatial visualization tools.
- Integrated spatial databases and Earth observation datasets into operational applications.

Student Researcher - Geospatial Deep Learning — **Jan 2023 - Dec 2023**
PhenoRob Lab, University of Bonn — Bonn, Germany

- Processed multi-temporal satellite and UAV datasets for environmental and agricultural analysis.
- Applied deep learning methods for spatial feature extraction and remote sensing applications.
- Performed preprocessing, validation, visualization, and interpretation of Earth observation datasets.

Senior GIS Analyst **Jul 2019 - Oct 2022**
Trimble Inc. Chennai, India

- Processed and validated large-scale geospatial datasets for mapping and infrastructure-related projects.
- Developed and supported QGIS plugin workflows for spatial data processing and quality control.
- Worked with raster and vector datasets including POI, AOI, and mapping layers.
- Performed data validation, quality checking, and structured GIS data management.
- Supported workflow automation using Python-based geospatial processing tools.

PROJECTS

Detecting Major Baltic Inflow Events and Baltic Sea Mass Variability Using GRACE(-FO) **2025-2026**
University of Bonn

- Investigated regional water mass variability in the Baltic Sea using GRACE and GRACE-FO satellite gravimetry.
- Integrated GRACE observations with satellite altimetry, ERA5 precipitation and runoff, GLDAS terrestrial water storage, GRDC river discharge, and Baltic Sea model outputs.
- Performed time-series analysis, anomaly detection, lag-correlation analysis, and event-based investigations of Major Baltic Inflow events.
- Developed MATLAB and Python workflows for large-scale environmental data processing and visualization.

Time Evolution of Winter Cooling and Associated Biogeochemical Processes in the Northern Arabian Sea **Jan 2019 - May 2019**
Research Project, National Institute of Oceanography (NIO), India
Supervisor: Dr. Jayu Narvekar

- Investigated winter cooling processes and their influence on physical and biogeochemical conditions in the Northern Arabian Sea.
- Analysed oceanographic and environmental datasets to study seasonal variability in sea surface temperature and related marine processes.
- Examined the relationship between winter cooling, ocean circulation, nutrient dynamics, and biological productivity.
- Developed skills in oceanographic data analysis, scientific interpretation, and environmental research methodologies.

High-Precision Space Geodesy Simulator **Apr 2024 - Apr 2025**

- Developed numerical simulation workflows for satellite trajectory modeling and geodetic analysis.
- Implemented efficient computation methods for high-precision space geodesy applications.

Mesoscale Eddy Detection in the Bay of Bengal **Aug 2018 - Apr 2019**
Bachelor's Thesis

- Investigated the occurrence and movement of mesoscale eddies in the Bay of Bengal using satellite altimetry observations.
- Processed sea surface height anomaly (SSHA) data to identify cyclonic and anticyclonic eddy structures.
- Analysed the spatial and temporal evolution of eddies and their propagation patterns within the study region.
- Evaluated the influence of mesoscale eddies on regional ocean circulation and sea surface dynamics.
- Developed geospatial workflows for data processing, visualization, and interpretation of oceanographic datasets.

LANGUAGES

German (A2), English (Fluent), Tamil (Native)